

Question Number	Answer	Additional guidance	Mark
5(a)(i)	• {location / position} of a {gene / allele} on a chromosome	ACCEPT chromatid for chromosome IGNORE DNA unqualified	(1)

Question Number	Answer	Additional guidance	Mark
5(a)(ii)	An explanation that makes reference to two of the following points: • different alleles {have a different DNA base sequence / code for different mRNA} (1) • which may result in a different {polypeptide / protein / pigment} being produced (1) • flies may inherit two different alleles / (different allele combinations) may result in {further / intermediate} {phenotypes / eye colours} (in addition to colours in table) (1)	REJECT in context of polygenic inheritance ACCEPT polypeptide has different {primary / secondary / tertiary structure} ACCEPT flies may be heterozygous ACCEPT combinations of different {alleles / pigments} result in {further / intermediate} {phenotypes / eye colours} (to what is in the table) ACCEPT both of the different alleles are expressed IGNORE polygenic inheritance / meiosis	(2)

Question Number	Answer	Additional guidance	Mark
5(b)	An explanation that makes reference to four of the following points: • post-transcriptional modification (during development of the embryo) / RNA splicing (1) • {introns / non-coding regions} (in Dsx pre-mRNA) removed by {enzymes / spliceosomes} (1) • {rearrangement of / removal of some} {exons / coding regions} (1) • translation occurs (of active mRNA) (1) • (resulting in) a different (Dsx) {primary sequence / sequence of amino acids / polypeptide} (1) • (Dsx-F) proteins result in female specific {cell development / structures / other proteins / cell modification} / converse for males (and Dsx-M)	ACCEPT different alleles for same gene ACCEPT alternative splicing IGNORE differential gene expression / genes being {switched off / expressed} IGNORE epigenetic modification ACCEPT introns are {spliced / removed} ACCEPT different {primary / secondary / tertiary} structure ACCEPT exons determine amino acid sequence ACCEPT one sequence of {exons / amino acids} results in Dsx-F, whereas another results in Dsx-M IGNORE proteins produced which modify the cell unqualified ACCEPT transcription factor produced which will activate gene(s) responsible for {male / female} development no ecf for genes being switched off	(4)

Question Number	Answer	Additional guidance	Mark
5(c)	An explanation that makes reference to the following points: • explanation of how a modification is caused (1) • correct effect of the histone modification (1) • methylation results in the gene being {switched off / silenced / not expressed / not transcribed} (1) • acetylation results in the gene being {switched on / expressed / activated / transcribed} (1) • (because) {RNA polymerase / transcription factors} can't {bind / access} (if gene is switched off) / converse (1)	e.g. methylation is addition of a {methyl / CH₃} group to {histone / lysine / arginine} e.g. acetylation is addition of an {acetyl / COCH₃} group to {histone / lysine} REJECT {methylation / acetylation} of {DNA / cytosine / CpG} e.g. histone methylation results in DNA being more tightly wrapped around histones / {histones / nucleosomes} closer together / heterochromatin / supercoiling} histone acetylation results in in the {histones / nucleosomes} being further apart / DNA less tightly wrapped around histones / euchromatin IGNORE DNA {more / less} coiled unqualified IGNORE distance between DNA strands IGNORE histones wrapping around DNA / histones becoming {loose / tighter} IGNORE less gene expression ACCEPT no {protein formed / protein synthesis} ACCEPT more gene expression ACCEPT protein formed / protein synthesis occurs REJECT DNA polymerase	(5)